

Credit Card Customer Segmentation

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Technologies: Python, Scikit-Learn, K-Means Clustering, PCA

1. Introduction

This project focuses on grouping credit card users into distinct segments based on their transactional habits and financial behavior. By analyzing customer profiles, financial institutions can design targeted marketing strategies, optimize credit limits, and enhance risk management frameworks. The project is implemented using unsupervised machine learning techniques, specifically K-Means Clustering and Principal Component Analysis (PCA).

2. Data Preprocessing & Cleaning

To ensure high data quality for spatial clustering, the following preprocessing workflow was applied to the raw dataset (CC_GENERAL.csv):

- **Missing Values Handling:** Structural null values found within the MINIMUM_PAYMENTS and CREDIT_LIMIT columns were imputed using the feature mean values to avoid data loss.
- **Feature Dropping:** The unique identifier column CUST_ID was removed since it contains no statistical variance or predictive value for grouping.
- **Feature Scaling:** Data was normalized using StandardScaler to scale features to a mean of 0 and a variance of 1. This prevents high-magnitude features from disproportionately dominating the distance calculations.

3. Model Training & Optimal Clusters

The K-Means clustering algorithm was configured and optimized through a systematic evaluation process:

- **The Elbow Method:** Within-Cluster Sum of Squares (WCSS) was tracked over an iteration range of 1 to 10 clusters to find the ideal mathematical inflection point.
- **Cluster Selection:** Based on the structural curve generated by the WCSS plot, an optimal value of 4 clusters was selected.
- **Label Assignment:** The finalized model applied the fit_predict() method to distribute cluster labels (0, 1, 2, 3) back into the primary customer dataframe.

4. Dimensionality Reduction & Visualization

Because the dataset consists of multiple dimensions, rendering a direct scatter plot on a 2D plane is impossible. To solve this, **Principal Component Analysis (PCA)** was used to reduce the dataset into

two principal orthogonal vectors (PCA1 and PCA2) that capture the maximum variance. The resulting components were used to build a clean 2D scatter plot visualization displaying the boundaries of each customer segment.

5. Final Cluster Insights

By extracting the group means for each cluster, the dataset reveals four distinct customer personas:

- **Cluster 0 (High Spenders):** Active users demonstrating a high volume of transactions and premium purchasing power.
- **Cluster 1 (Low Spenders):** Conservative users who interact with their credit cards minimally and maintain low spending baselines.
- **Cluster 2 (Frequent Users):** Customers utilizing credit as a standard utility for recurring, small-scale daily purchases.
- **Cluster 3 (Premium Customers):** VIP tier cardholders characterized by exceptionally high credit limits and substantial structural balances.

6. Conclusion

Developed by team **Powerpuffs**, this end-to-end data pipeline successfully turns unstructured credit card metadata into actionable business intelligence. The resulting model can easily be integrated into corporate marketing stacks to trigger localized rewards plans and targeted financial products.